PCEOT402
ELECTRICAL MACHINES

Module 1
DC Machines

DC motor – back emf – torque equation – numerical problems (1 hour)
Classification of DC motors – steady-state equations – numerical problems (1 hour)
Characteristics of shunt and series motors (1 hour)
Starting of DC motors (concepts only)
Power flow diagram – losses and efficiency – numerical problems (1hour)
Testing - Swinburne’s test – numerical problems (1 hour)



DC Motors

DC Motor

• DC motor is a machine which converts Electrical energy into Mechanical energy(DC Input)

•These **motors** are used to drive generators, fans and pumps.

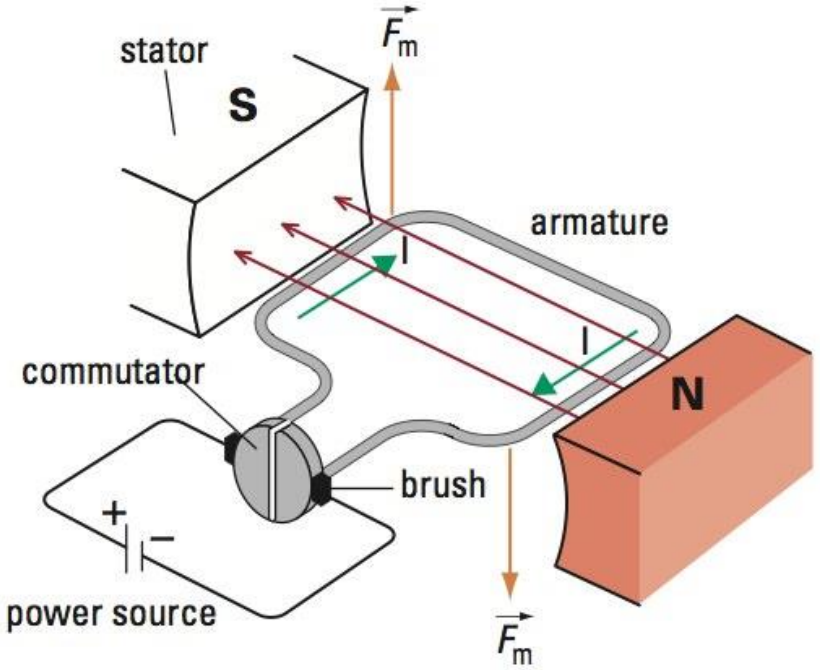


Figure 12.28 A simple DC motor

Principle of Operation

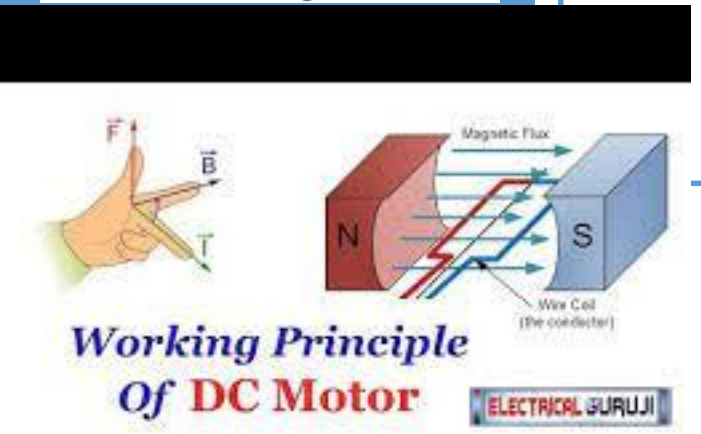
$$F = BIl$$

Where:

B = Flux Density

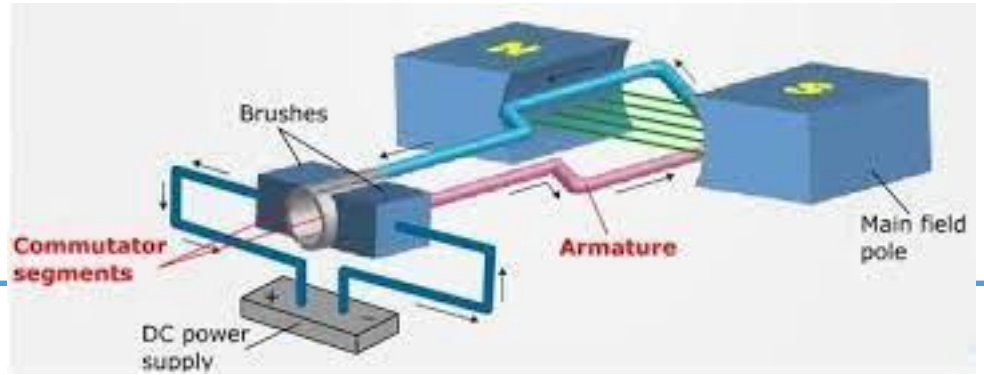
I = Current

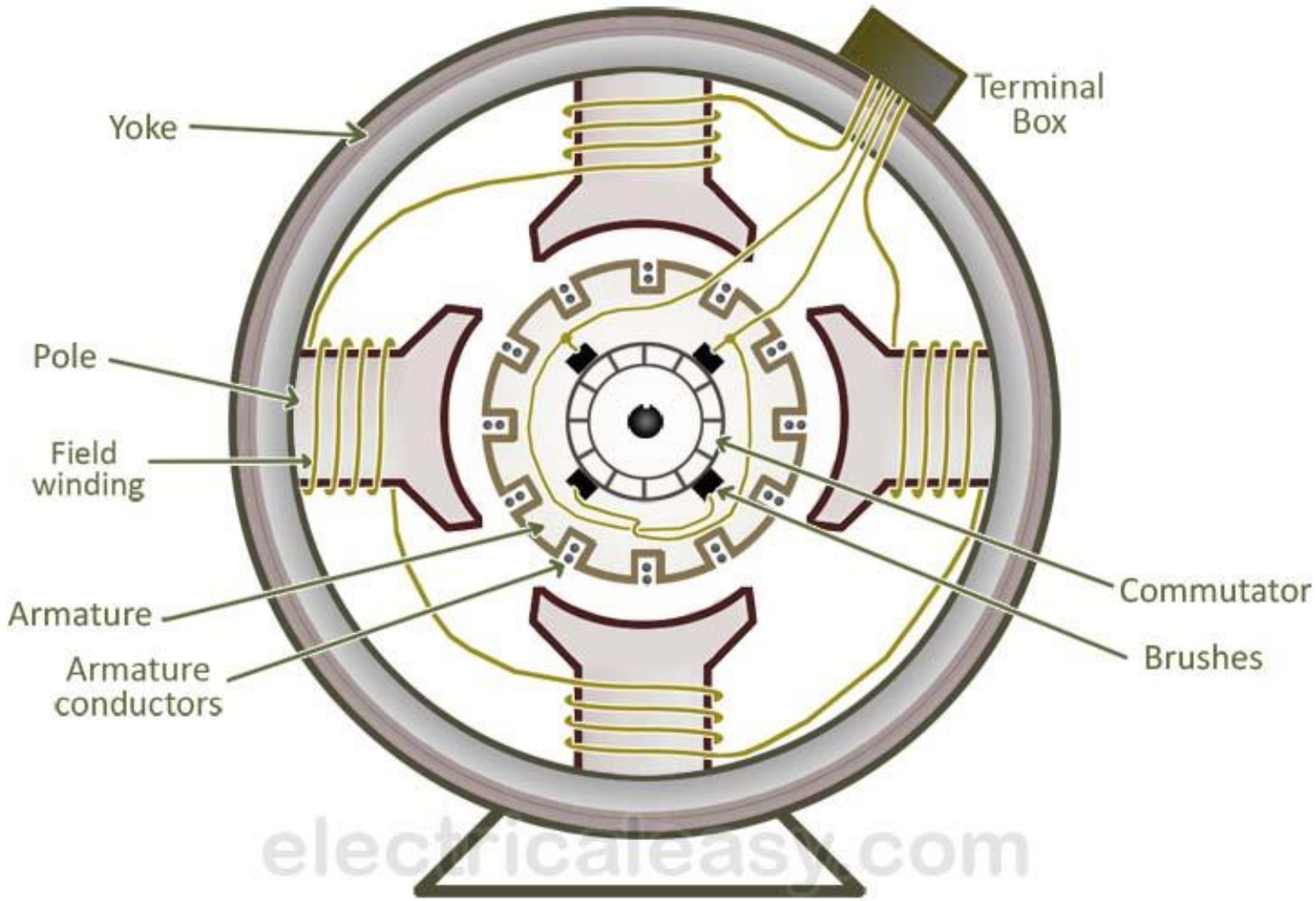
l = Length



DC motor operates on the principle that, a mechanical force acts on a current carrying conductor placed in a magnetic field.

If the conductor is free to move, it moves in the direction of force, which is determined by Flemings Left Hand rule.



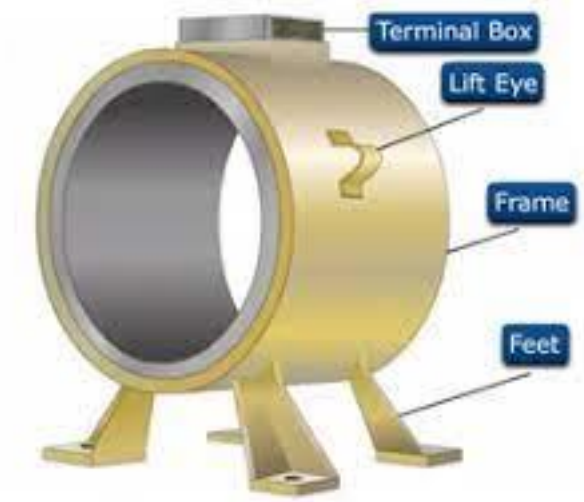

Yoke
Terminal Box
Pole
Field winding
Armature
Armature conductors
Commutator
Brushes
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Construction of DC Motor

Magnetic Field System (Stator)

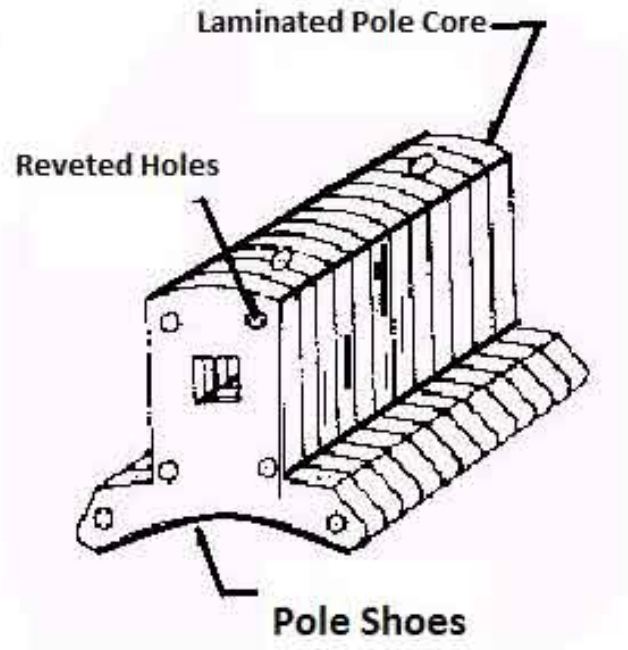
Yoke

1. Protects the machine (Frame)
2. Mechanical support to poles
3. Low reluctance path
4. Cast Iron/ Cast Steel/ Rolled Steel



Construction of DC Motor

Magnetic Field System (Stator)



Pole Core

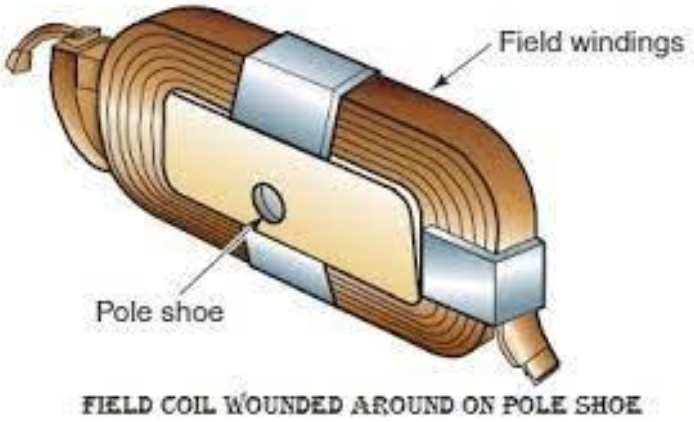
1. **Pole core** acts as electromagnet with field winding wound to produce flux when excited
2. **Pole core** is laminated to reduce Eddy Current
3. **Pole core** offers low reluctance path
4. **Pole core** material used can be Cast Iron/ Cast Steel/ Rolled Steel

Pole Shoe

1. **Pole shoe** supports field coils
2. **Pole shoe** increases the cross sectional area of magnetic circuit
3. **Pole shoe** is laminated to reduce Eddy Current
4. **Pole shoe** offers low reluctance path
5. **Pole shoe** material used can be Cast Iron/ Cast Steel/ Rolled Steel

Construction of DC Motor

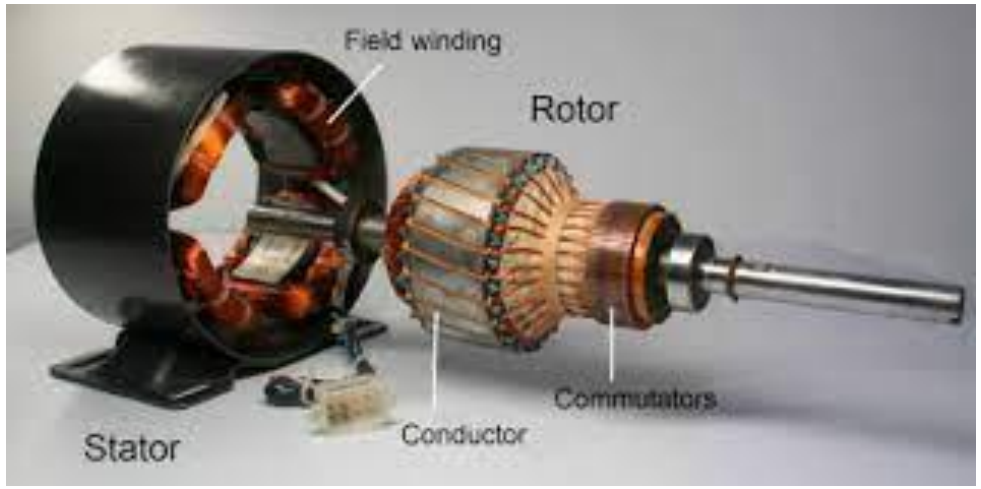
Magnetic Field System (Stator)



FIELD COIL WOUNDED AROUND ON POLE SHOE

Field Windings

1. Connected around the pole shoe.
2. Pole direction is found by using Maxwell's right hand grip rule.
3. Copper is preferred than Aluminum for bending easily.
4. Forms as the electromagnet as it is excited.





DC motor stator with poles visible.

Construction of DC Motor

Armature (Rotor)

Armature Core

1. Slots to accommodate armature windings. (Conductors)
2. Lamination to reduce Eddy Current Loss
3. Low reluctance path
4. Cast Iron/ Cast Steel/ Rolled Steel





Construction of DC Motor

Armature (Rotor)

Armature Windings

1. Armature conductors are connected to form armature windings
2. Voltage/ Magnetic Flux is induced when conductor is moved in magnetic field for generator and motor respectively.
3. Connection taken to external circuit
4. Copper for better conductivity







Construction of DC Motor

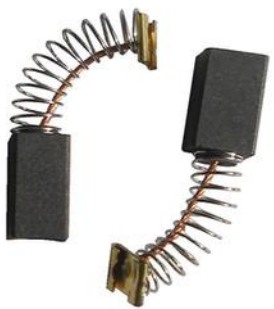
Commutator & Brushes



Commutator

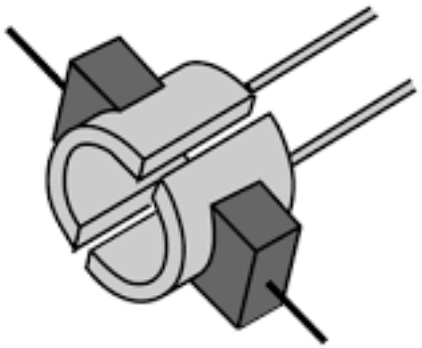
1. Mechanical rectifier which collects current from brushes and supply it to armature conductors
2. Copper segments with mica insulation in between.
3. Provides Unidirectional Torque.

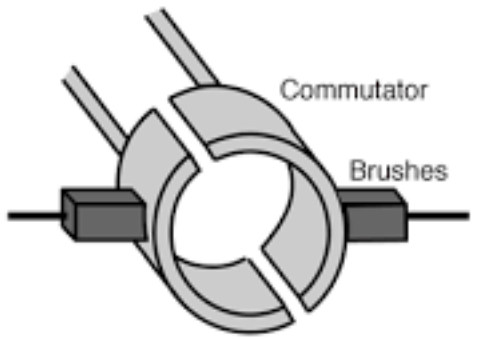




Brushes

1. Gives Current from to armature conductors through commutator
2. Carbon or Graphite
3. Wearing can happen with time.





Construction of DC Motor

Shaft

Shaft

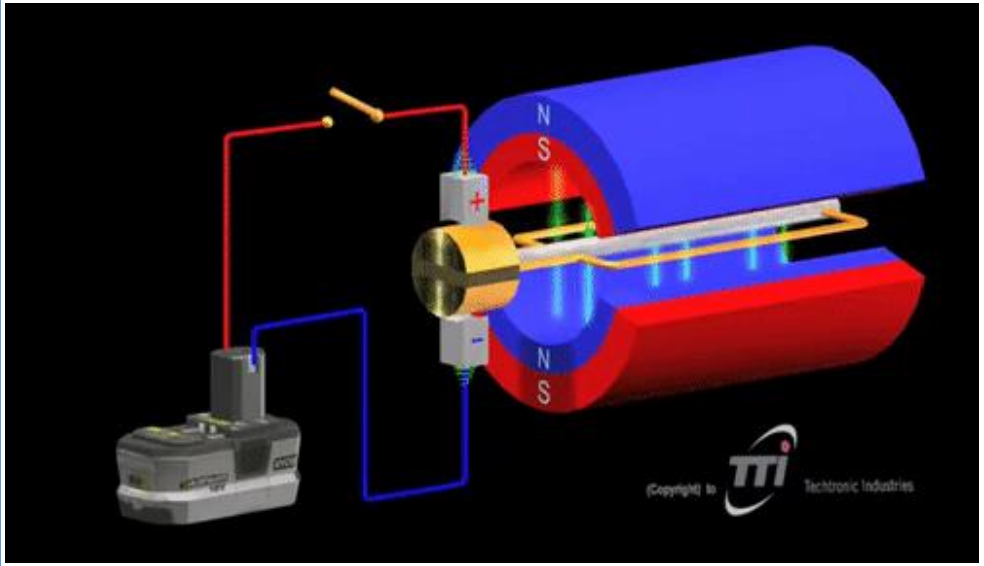
1. To transfer mechanical power from the machine to the load.
2. Attached to rotating parts like armature core, commutator, cooling fans etc.
3. Mild steel in case of low applications and Nickel, Nickel- Chromium in case of heavier applications.

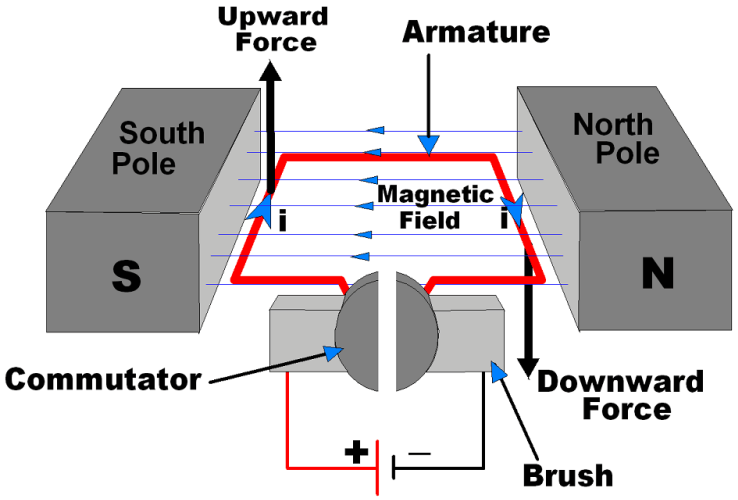


Working of DC Motor

Working

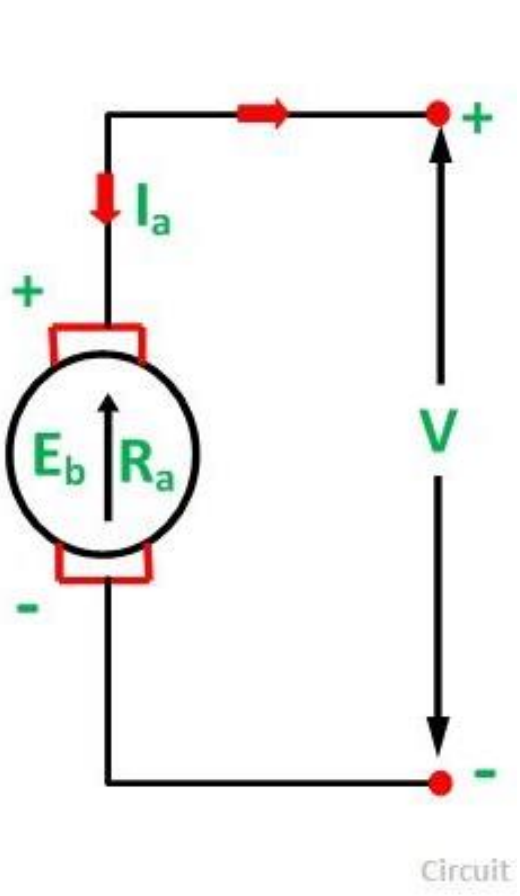
- When armature windings are connected to a DC supply, an electric current sets up in the winding.
- Magnetic field may be provided by field winding (electromagnetism) or by using permanent magnet
- In this case, current carrying armature conductors experience a force due to the magnetic field.
- Commutator is made segmented to achieve unidirectional torque.
- Otherwise, the direction of force would have reversed every time when the direction of movement of conductor is reversed in the magnetic field.
- **This is how a DC motor works.**





DC Motor Conceptual Diagram

Back EMF



• When the armature of a motor is rotating, the conductors are also cutting the magnetic flux lines and hence according to the Faradays law of Electromagnetic Induction, an EMF induces in the armature conductors.
• The direction of this induced EMF is such that it opposes the armature current (I_a)(Lenz's Law)
• Thus this EMF is known as Back EMF(Eb).
•Magnitude of the **Back EMF** can be given by EMF equation of a generator.

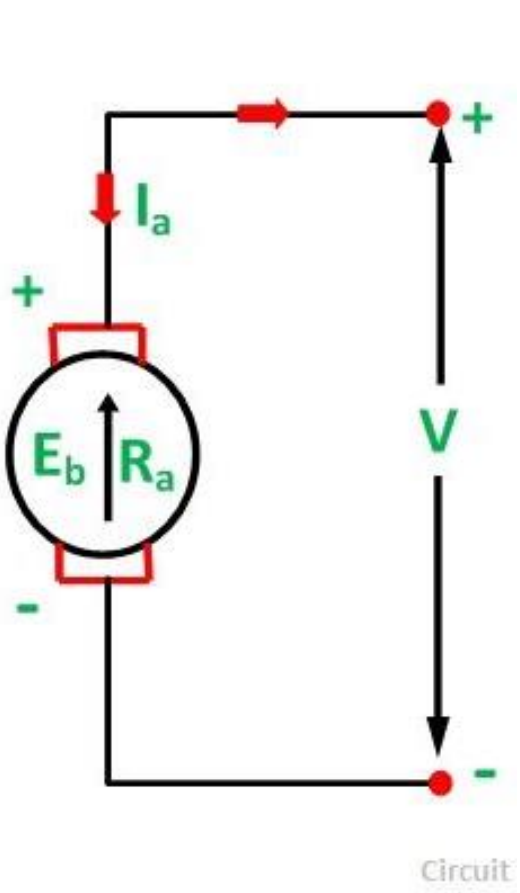
$$E_b = \frac{\phi ZNP}{60A}, Volts$$

• Magnitude of back EMF is directly proportional to speed of the motor.

Significance of Back EMF

• Consider the load on a dc motor is suddenly reduced.
• In this case, required torque will be small as compared to the current torque.
• Speed of the motor will start increasing due to the excess torque.
• Hence, being proportional to the speed, magnitude of the back EMF will also increase.
• With increasing back EMF armature current will start decreasing.
• Torque being proportional to the armature current, will decrease until it becomes sufficient .
•Thus, speed of the motor will regulate.
• On the other hand, if a dc motor is suddenly loaded, the load will cause decrease in the speed.
• Due to decrease in speed, back EMF will also decrease allowing more armature current.
• Increased armature current will increase the torque to satisfy the load requirement.
• Hence, presence of the **back EMF makes a dc motor 'self-regulating'**.

Voltage and Power Equations



- **Voltage Equation of a DC Motor**

$$V = E_b + I_a R_a$$

$$V = Applied_Voltage$$

$$E_b = Back_EMF$$

$$I_a = Armature_Current$$

$$R_a = Armature_resistance$$

Factors affecting Speed

$$E_b = \frac{\phi ZNP}{60A}$$

$$N = \frac{E_b 60A}{\phi PZ}$$

$$P, Z, and A - Fixed$$

$$N \propto \frac{E_b}{\phi}$$

Mechanical Power delivered

$$E_b = V - I_a R_a$$

$$Power = E_b I_a, \therefore E_b I_a = VI_a - I_a{}^2 R_a$$

$$VI_a = Power'supplied'to'armature$$

$$I_a{}^2 R_a = Armature'copper''loss$$

$$\therefore P_m = E_b I_a, Watts$$

Torque Equation

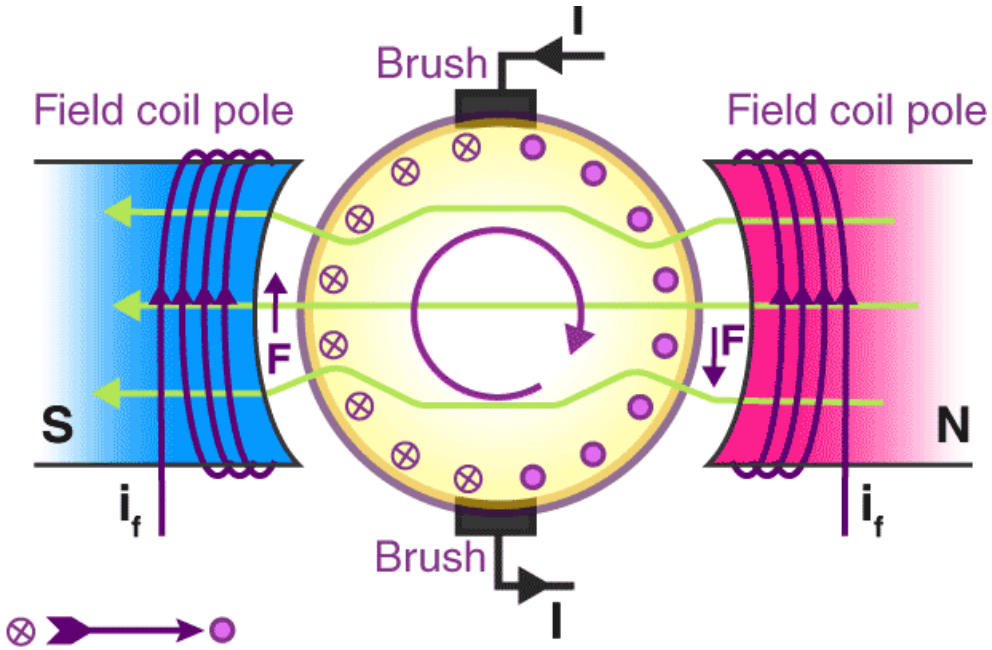
Production of Torque in a DC Machine

•When field magnets are excited and current is sent through the armature conductors, force is experienced to rotate the armature.

• Applying Flemings Left Hand rule in the below figure, the force acting on each conductor is found to be in clockwise direction (conductors near N pole has current moving out from the plain and conductors near S pole has current moving into the plain)

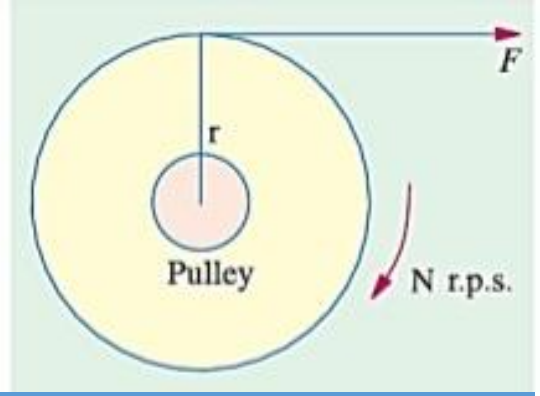
• These forces in armature conductor add up to produce the driving torque which causes rotation of the armature.

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Production of torque in a DC motor

Torque Equation



Torque Equation

- Pa= Power developed by armature
- Ta = electromagnetic Torque developed in N-M
- Eb = back emf produced
- Z= Total number of conductors
- B= Flux Density(Wb/m)=
- i= Current in each conductor=Ia/A, Ia= Armature Current, A= Number of parallel paths

$$Pa = Torque * angularvelocity$$

$$P_a = T\omega;$$

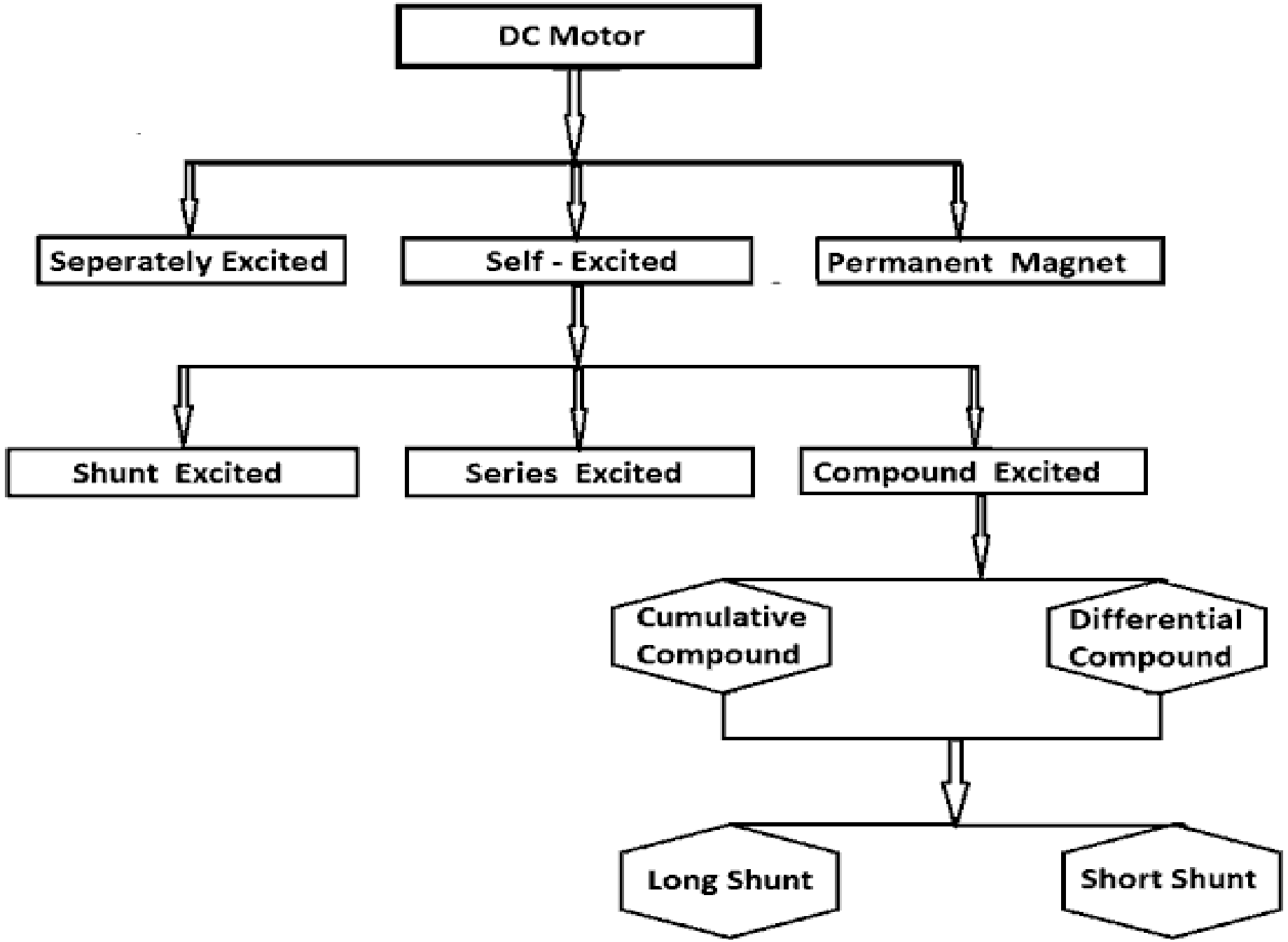
$$\omega = \frac{2\prod N}{60};$$

$$E_b I_a = T\frac{2\prod N}{60}$$

$$T_a = Z\frac{\phi P}{2\pi}\frac{I_a}{A}(N-m)$$

$$T_a = 0.159\phi Z I_a(\frac{P}{A})(N-m)$$

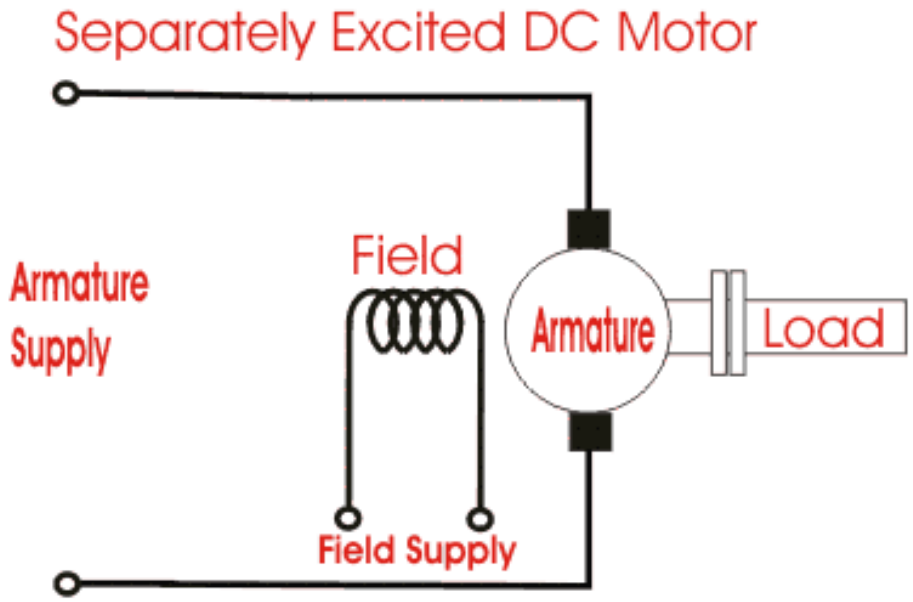
$$T\alpha\phi I_a$$


DC Motor
Seperately Excited
Self - Excited
Permanent Magnet
Shunt Excited
Series Excited
Compound Excited
Cumulative Compound
Differential Compound
Long Shunt
Short Shunt

Types of DC Motors

Separately Excited

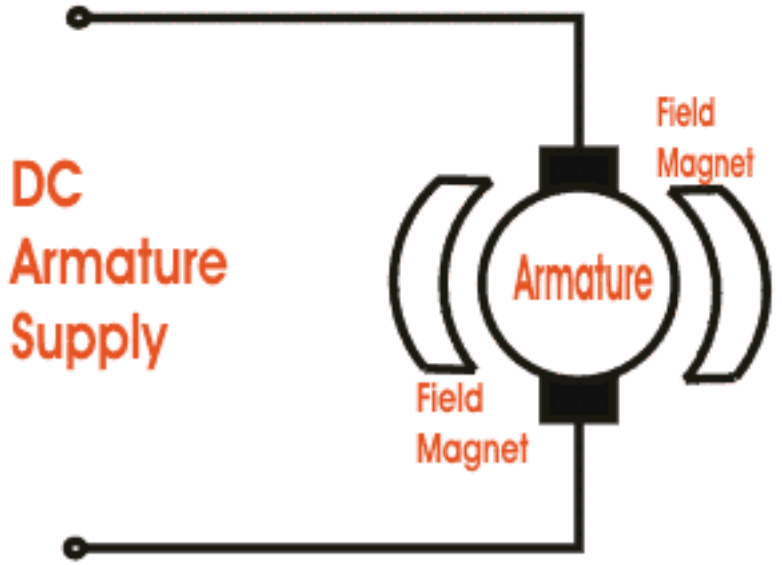
•Field magnet winding and armature conductor windings are supplied from an independent external DC source or battery.



Types of DC Motors

Permanent Magnet

•Permanent magnets are used instead of Field wound electromagnets.



Permanent Magnet DC Motor

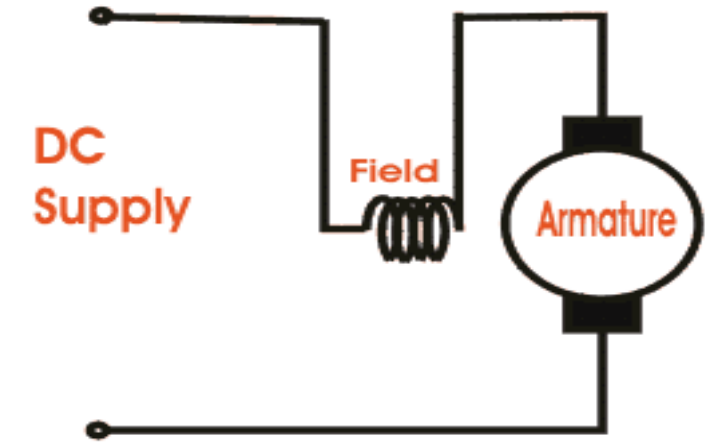
Types of DC Motors

Self Excited

•A DC Motor whose field winding is connected either in series or in parallel or partly in series, partly in parallel to the armature winding.
•There are three types of self-excited motors depending upon the manner in which the field winding is connected to the armature, namely;
(i) Series Motor
(ii) Shunt Motor
(iii) Compound Motor

1. Series Motor

•In a series wound motor has the field winding connected in series with the armature.
•The whole current applied to the motor passes through the field windings as well as the armature.



Series Excited DC Motor

$$V = E_b + I_a(R_a + R_{se})$$

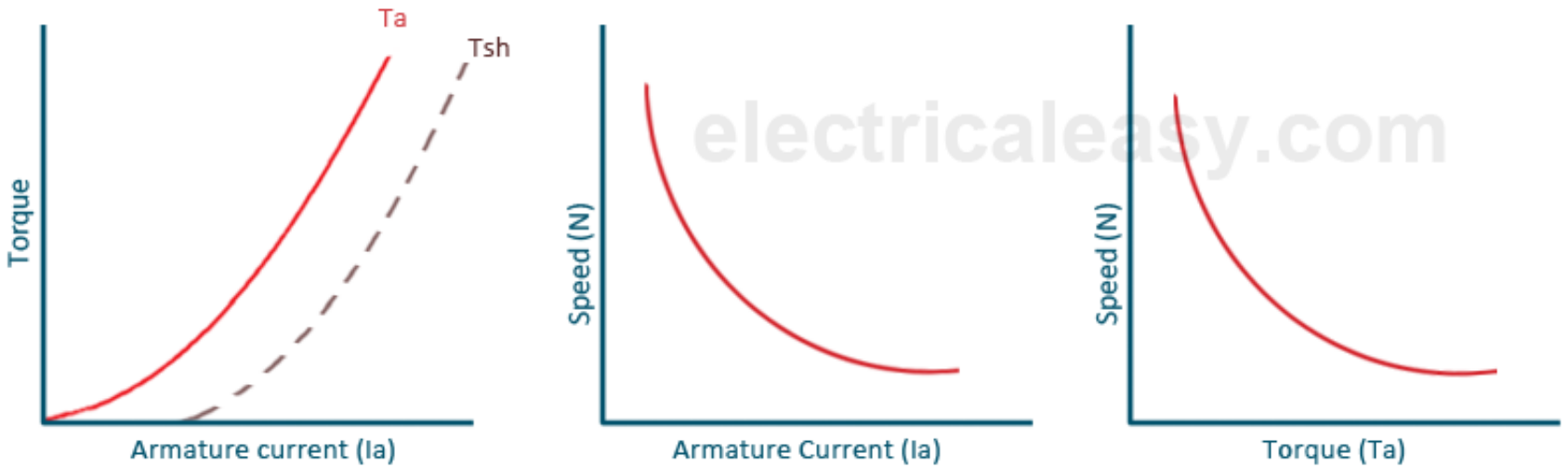
Types of DC Motors

Performance Characteristics of Series Motors

- As armature current increases with load, the speed comes down sharply.
- Speed is dangerously high at no load.

Applications of Series Motors

- Heavy duty applications like electric railways, rolling mills, cranes etc.

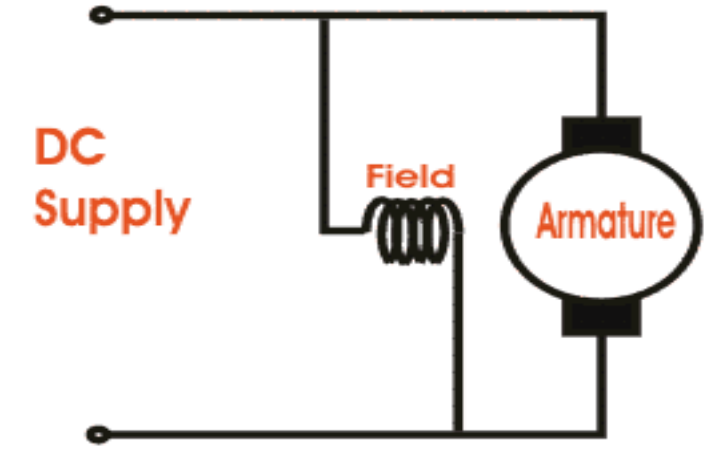


Characteristics of DC series motor

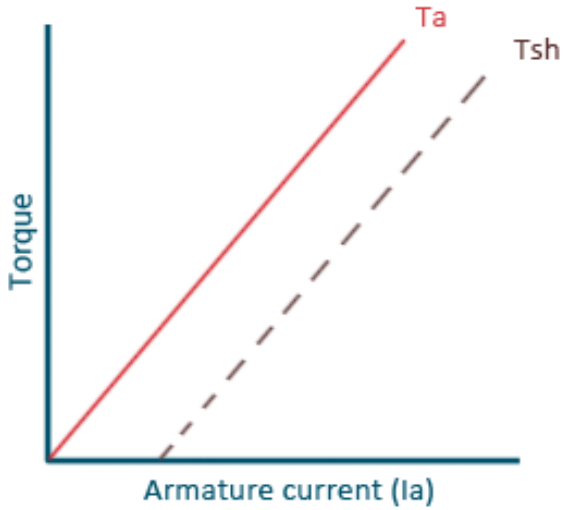
Types of DC Motors

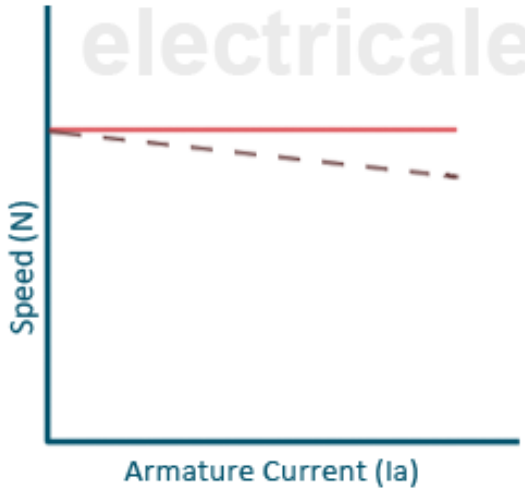
2. Shunt Motor

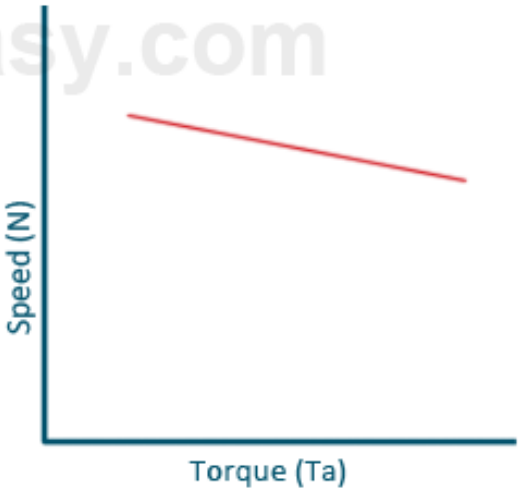
•Field winding is connected in parallel with the armature.
•The current is offered in two paths, one through armature and the other through field coils.



Shunt Excited DC Motor







electricaleasy.com

$$V = E_b + I_a R_a$$

Characteristics of DC shunt motor

Performance Characteristics of DC Shunt Motor

•Medium starting torque.
•Constant Speed Motor(Speed does not vary as load is increasing)

Applications of Shunt Motors

•Driving constant speed line shafts, vacuum cleaners, compressors, etc.

Types of DC Motors

3. Compound Motor

•Field winding is connected in parallel and series with armature

•In Cumulative compound $\phi_{Total} = \phi_{Series} + \phi_{Shunt}$

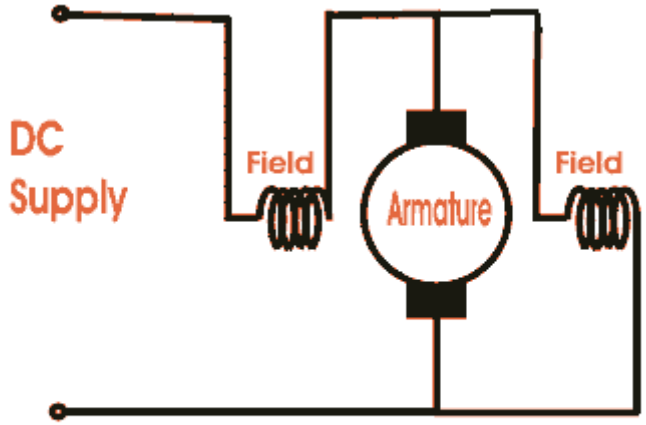
•In Differential Compound $\phi_{Total} = \phi_{Series} - \phi_{Shunt}$

a. Short Shunt DC Motor

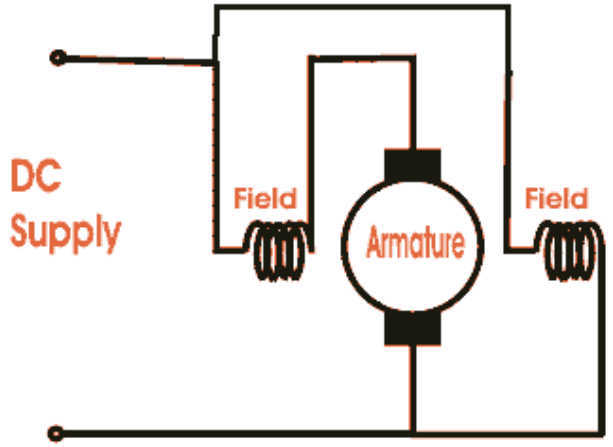
•Shunt field winding is only parallel to the armature winding and not the series field winding.

b. Long Shunt DC Motor

•Shunt field winding is parallel to both the armature winding and the series field



Short Shund DC Motor



Long Shunt DC Motor

Questions

1. A 250 V, DC shunt motor takes 41 amperes at full load. Resistance o motor armature and shunt field windings are 0.1 ohm and 250 respectively. Find the back emf on full load?

The armature of a 4 pole dc shunt motor has a lap winding accommodated in 60 slots each containing 20 conductors. If the useful flux per pole is 23m Wb, calculate the total torque developed in Newton meters when the armature current is 50 amperes?

A 4 pole, 250 volt DC series motor has a wave connected armature with 1254 conductors. The flux/pole is 22m Wb when the motor is taking 50 ampere. Armature resistance is 0.2 ohms and series field resistance is 0.2 ohms. Calculate the speed?

STARTING OF DC MOTOR

NECESSITY OF STARTER

The equation of armature current in a DC shunt motor is written as

$$I_a = \frac{V - E_b}{R_a}$$

$$E_b \propto N$$

At the time of starting, speed is zero (motor is yet to pick up its speed) and hence back emf is zero. Therefore,

$$I_a = \frac{V}{R_a}$$

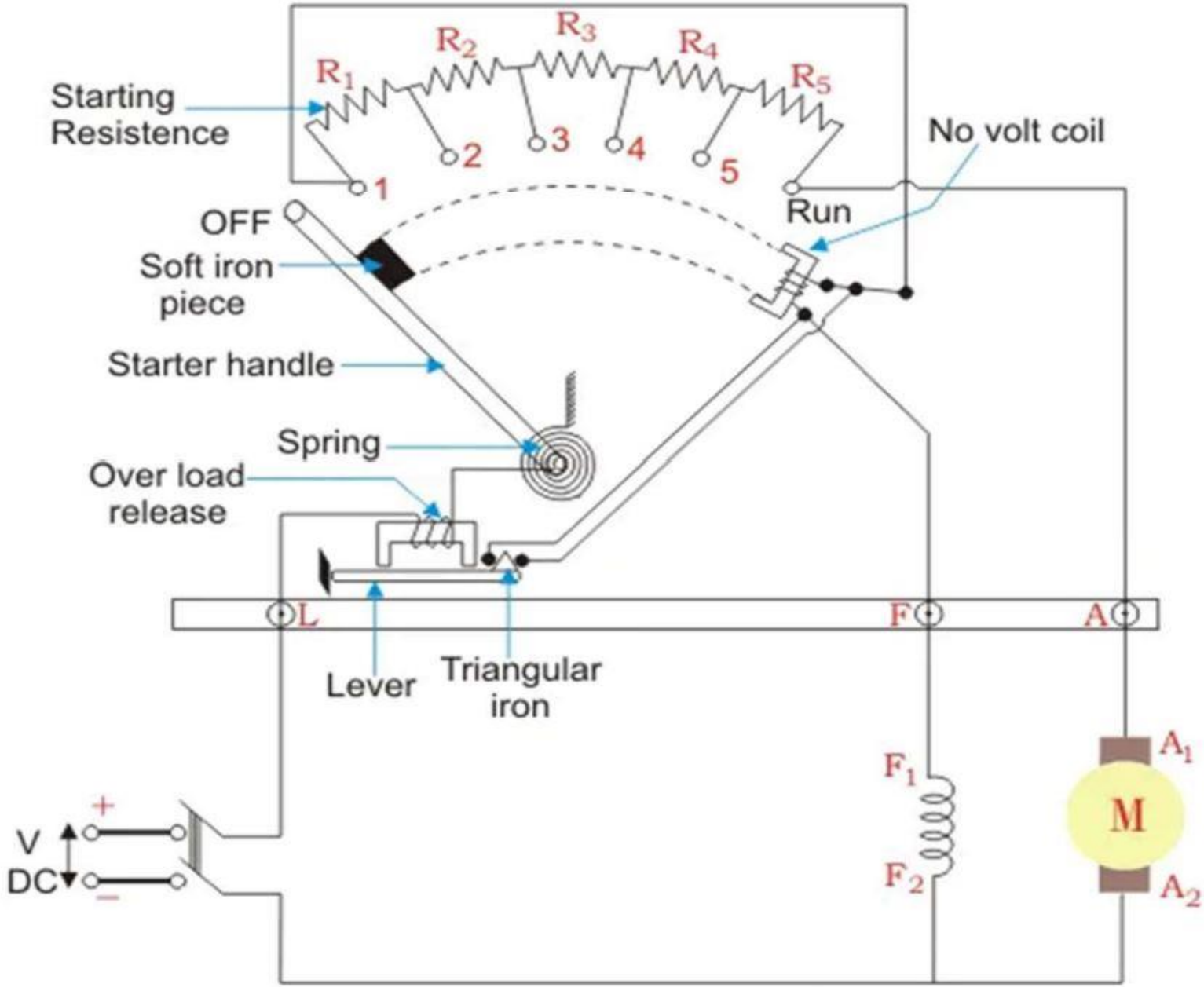
For a practical DC motor, armature resistance is very low, generally about 0.5Ω. Hence, large current flows through the armature during starting.

For example, a 10kW, 250V shunt motor may have $R_a \approx 0.2\Omega$. If switched on directly, its starting armature current would be 250/0.2 = 1250A whereas the rated armature current is only 40A.

Such heavy inrush of starting current may result in

i) Detrimental sparking at the commuatator
ii) Damage to the armature winding and deterioration of the insulation due to overheating
iii) High starting torque and quick acceleration – which may damage the rotating parts of the motor and the load and
iv) Large dips in the supply voltage.

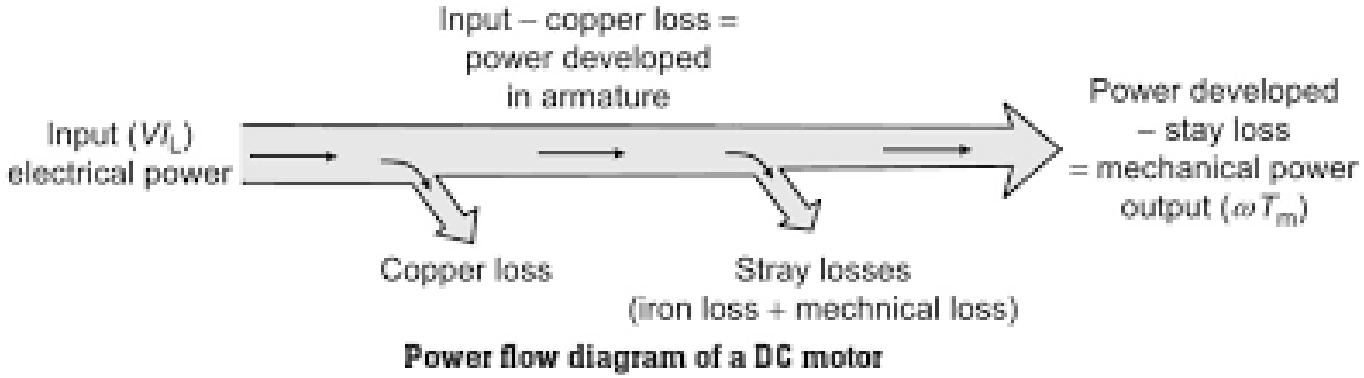
Activate Win
Go to Settings to



Three Point Starter

Losses in a DC Motor

- As total input is not received at output of DC motor, its understood that losses happen when electromechanical energy conversion is happening



Power flow diagram of a DC motor

a. **Copper Losses**

•All windings have some resistance and hence there are copper losses associated with current flow in them.

•They generate heat proportional to the resistance of the conductor.

•Copper losses can be divided into

1. Armature Copper Loss(Copper loss occurring due to resistance of armature winding)
   - This loss is about 20%-30 % of full load losses.
   - Armature Copper loss is variable.
2. Shunt Copper Loss(Copper loss occurring due to resistance of shunt field winding)
3. Series Copper Loss(Copper loss occurring due to resistance of series field winding)
   - Both these loss is about 30%-40 % of full load losses.
   - Field Copper losses are constant.

$$Armatureloss = I_a^{\,2} R_a$$

$$Shuntloss = I_{sh}^{\,2} R_{sh}$$

$$Seriesloss = I_{se}^{\,2} R_{se}$$

Losses in a DC Motor

b. Mechanical Losses

•Losses due to mechanical friction of the machine.

•Mechanical losses can be divided into

1. Friction Loss(Due to bearing and brushes)

This loss is about 20%-30 % of full load losses.

2. Windage loss(Due to friction caused by air circulated for cooling purpose)

c. Core Losses(Iron Losses)

•Due to rotating armature in the magnetic field.

•Core losses can be divided into

1. Hysteresis Loss(Heat loss due to magnetic properties of the Armature)

This is due to the reversal of magnetization of the armature core.

$$W_h = \eta B^{1.6}_{\max} fV(Watts)$$

2. Eddy Current Loss(Due to the Current generated by cutting magnetic flux by armature conductors)

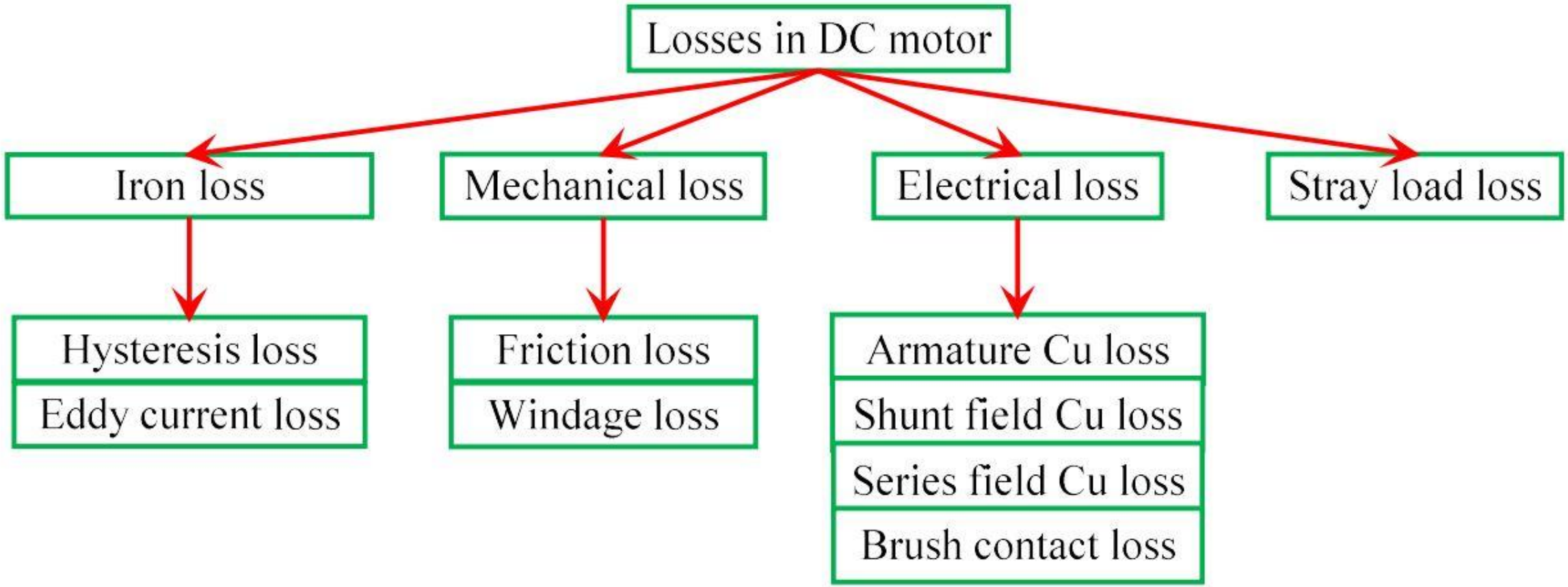
Current produced due to the EMF generated is called Eddy Current.

Power loss happens in the form of heat

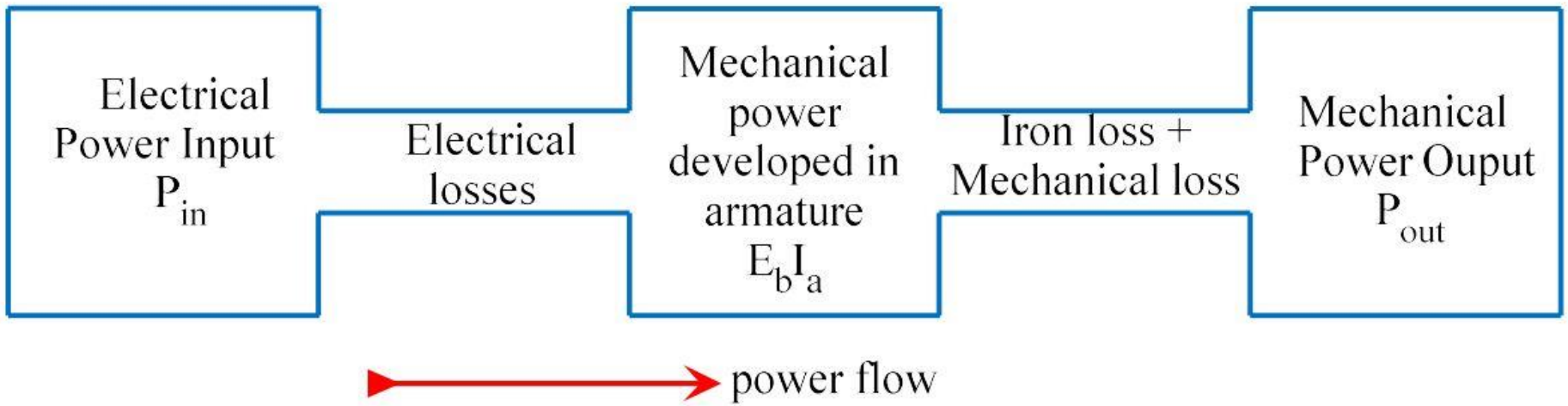
$$W_e = kB^2_{\max} f^2 t^2 V(Watts)$$

Total Loss= Variable armature Copper loss+ Constant Loss(Wc)

LOSSES & EFFICIENCY



POWER FLOW DIAGRAM



The following are the three efficiencies calculated for DC motors:

i) Electrical efficiency $\eta_e = \dfrac{\text{Mechanical power developed}}{\text{Electrical power input}}$

ii) Mechanical efficiency $\eta_m = \dfrac{\text{Mechanical power output}}{\text{Mechanical power developed}}$

iii) Overall efficiency $\eta = \dfrac{\text{Mechanical power output}}{\text{Electrical power input}}$

$$\eta = \eta_e \times \eta_m$$

CONDITION FOR MAXIMUM EFFICIENCY

For a shunt motor, assuming $I_L = I_a$,

$\eta = \dfrac{VI_L - I_L^2 R_a - W_c}{VI_L}$ where W_c is the constant loss

For maximum efficiency, $\dfrac{d\eta}{dI_L} = 0$

$$VI_L \times (V - 2I_L R_a) - (VI_L - I_L^2 R_a - W_c) \times V = 0$$

$$VI_L - 2I_L^2 R_a - VI_L + I_L^2 R_a + W_c = 0$$

$$I_L^2 R_a = W_c$$

For a shunt motor, efficiency is maximum, when variable loss is equal to constant loss, ie. $I_a^2 R_a = W_c$

For compound motor, efficiency is maximum when $I_a^2 (R_a + R_{se}) = W_c$

The line current corresponding to maximum efficiency, $I_L = \sqrt{\dfrac{W_c}{R_a}}$

Testing of DC Motor

- To perform the electric characteristics of a DC machine after construction, testing is to be done.
- Testing is used for finding Losses, temperature rise and to pre determine the efficiency of DC machine.
- The Important Tests of DC machine are

1. Swinburne's Test(Indirect Test)
2. Hopkinsons Test(Regenerative Test)
3. Retardation Test(Running Down Test)

- Losses in a DC machine are,

1. Copper Losses(Armature Copper Loss, Field Copper Loss and Brush loss)
2. Iron Losses/ Core Losses(Hysteresis Loss and Eddy Current Loss)
3. Mechanical Loss(Windage loss and Frictional Loss)
4. Stray Losses(Miscellaneous Losses)

Copper Losses:

a) Armature Copper Losses/ Variable Loss $P_v = I_a{}^2 R_a$

b) Field Copper Losses, $P_f = I_{sh/se}{}^2 R_{sh/se}$

Core Loss

a) No load Core loss, P_{i0}

Mechanical Losses

a) Windage and Friction Losses, P_{wf}

$$\text{Motor Input at no Load } (V_a I_{ao}) = P_{i0} + P_{wf} + I_{ao}{}^2 R_a$$

$$\text{Rotational Loss } (P_{i0} + P_{wf}) = V_a I_{ao} - I_{ao}{}^2 R_a$$

$$\text{Constant Loss } (P_k) = (V_a I_{ao} - I_{ao}{}^2 R_a) + I_f{}^2 R_f$$

$$\text{Total Loss} = \text{Constant Loss} + \text{Variable Loss}$$

Testing of DC Motor

A. Swinburne's Test

- In this method, the D.C Motor run at no load and losses of the machine are determined.
- Once the losses of the machine are known, its efficiency at any desired load can be determined in advance.
- It may be noted that this method is applicable to those machines in which flux is practically constant

at all loads e.g., shunt and compound machines.

The test insists of two steps:

1. **Determination of hot resistances of windings**

- The armature resistance and shunt field resistance are measured using a battery, voltmeter and ammeter.
- Since these resistances are measured when the machine is cold(15°C), they must be converted to values corresponding to the temperature at which the machine would work on full-load.
- Generally, these values are measured for a temperature rise of 40°C above the room temperature
- Let the hot resistances of armature and shunt field be R_a and R_{sh} respectively.

Then, the 'hot' resistance, allowing a temperature rise of 50°C is found thus :

$$R_{15} = R_0(1 + 15\alpha_0)\ ;\ R_{65} = (1 + 65\alpha_0),\ R_{65} = R_{15} \times \frac{1 + 65\alpha_0}{1 + 15\alpha_0}$$

Taking $\alpha_0 = 1/234.5$, we have $\quad R_{65} = R_{15} \times \frac{234.5 + 65}{234.5 + 15} = 1.2\ R_{15}$ (approx.*)

Testing of DC Motor

2. Determination of constant losses

- The machine is run as a motor on no-load with supply voltage adjusted to the rated voltage
- The speed of the motor is adjusted to the rated speed with the help of field regulator R as shown is Fig: 1.

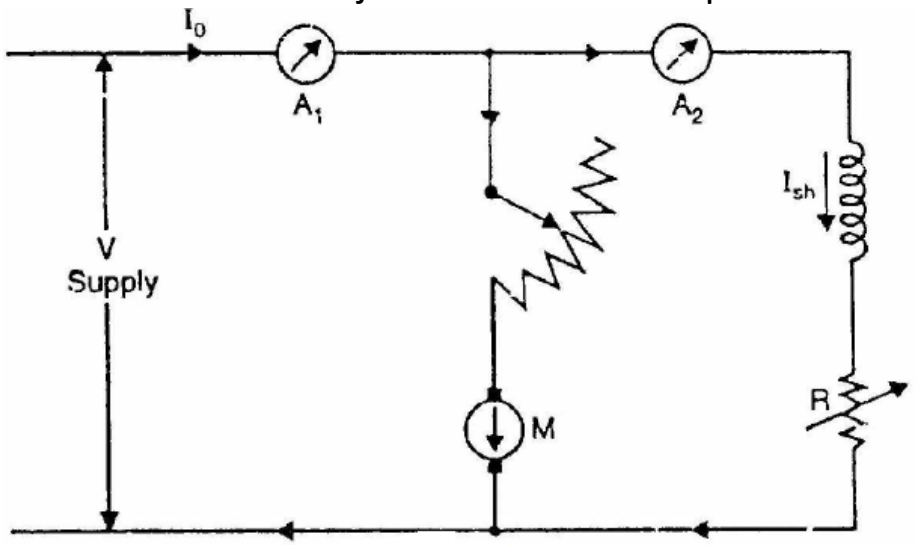


Fig: 1- Swineburn's Test

$V = \text{Supply Voltage}$

$I_0 = \text{No Load Current}$

$I_{sh} = \text{Shunt Field Current}$

$\text{No Load Armature Current}(I_{a0}) = I_0 - I_{sh}$

$\text{No Load input power to motor} = VI_0$

$\text{No Load input power to Armature} = VI_{a0}$

$= V(I_0 - I_{sh})$

•Since the input power doesnt supply load, it is used to overcome the losses such as

1. Iron Loss (Core Loss)
2. Friction Loss (Constant Loss)
3. Windage Loss (Constant Loss)
4. Armature Copper Loss

$I_{a0}^2 R_a = (I_0 - I_{sh})^2 R_a$

Constant Losses (Wc)= Input –Armature Copper Losses=

$VI_0 - (I_0 - I_{sh})^2 R_a$

Testing of DC Motor

Efficiency when running as a motor

Input power to motor = VI

Armature Cu loss = $I_a^2 R_a = (I - I_{sh})^2 R_a$

Constant losses = W_C found above

Total losses = $(I - I_{sh})^2 R_a + W_C$

$\therefore$ Motor efficiency, $\eta_m = \frac{\text{Input} - \text{Losses}}{\text{Input}} = \frac{VI - (I - I_{sh})^2 R_a - W_C}{VI}$

Advantages of Swinburne's test

(i) The power required to carry out the test is small because it is a no-load test. Therefore, this method is quite economical.

(ii) The efficiency can be determined at any load because constant losses are known.

(iii) This test is very convenient.

Disadvantages of Swinburne's test

(i) It does not take into account the stray load losses that occur when the machine is loaded.

(ii) This test does not enable us to check the performance of the machine on full-load. For example, it does not indicate whether commutation on full-load is satisfactory and whether the temperature rise is within the specified limits.

(iii) This test does not give quite accurate efficiency of the machine. It is because iron losses under actual load are greater than those measured. This is mainly due to armature reaction distorting the field.

Testing of DC Motor

Problem:

Example 31.3. *A 220 V, d.c. shunt motor at no load takes a current of 2.5 A. The resistances of the armature and shunt field are 0.8 Ω and 200 Ω respectively. Estimate the efficiency of the motor when the input current is 20 A. State precisely the assumptions made.*

(Electrical Technology, Kerala Univ. 1986)

Solution. No-load input = $220 \times 2.5 = 550$ W

This input meets all kinds of no-load losses *i.e.* armature Cu loss and constant losses.

$I_{sh} = 220/200 = 1.1$ A. No-load arm current, $I_{a0} = 2.5 - 1.1 = 1.4$ A

No-load armature Cu loss = $I_{a0}^2 R_a = 1.4^2 \times 0.8 = 1.6$ W

Constant losses = $550 - 1.6 = 548.4$ W

When input current is 20 A

$I_a = 32 - 1.1 = 30.9$ A ; Armature Cu loss = $30.9^2 \times 0.8 = 764$ W

Total loss = $764 + 548.4 = 1312$ W (approx.) ; Input = $220 \times 20 = 4{,}400$ W

Output = $4{,}400 - 1{,}312 = 3{,}088$ W ; Efficiency = $(3088/4400) \times 100 =$ **70.2%**

In the above calculations, it has been assumed that :

1. mechanical losses remain constant even through motor speed changes from no-load to the given load.
2. effect of armature reaction on main pole flux with a consequent change in iron losses has been neglected.
3. decrease in flux due to increase in shunt resistance by heating has been neglected.

Thank you

Insert the title of your subtitle Here

